Large Disproportionate Mortality Effects on the Global Poor Drive a Higher Income-Weighted Social Cost of CO_2

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Income-weighting is an important concept in benefit-cost analysis, and it has been implemented by multiple countries around the world. The income-weighted social cost of carbon (SC-CO_2) reflects the monetized effect of emitting one tonne of CO_2 on well-being by applying larger weights to dollar-denominated damages in lower-income areas than higher-income areas, consistent with evidence on the declining marginal well-being of income. In this study, we combine a state-of-the-art climate-economy model with sector-specific damage functions and a temperature-related mortality damage function that accounts for adaptation to calculate an income-weighted SC-CO_2. We find that climate change will cause an estimated 1.6 million temperature-related deaths per year in 2100 (-1.5M–5.6M: 5%–95% range) and 85 million total temperature-related deaths from 2023–2100 (-20M–225M: 5%–95% range). The hottest and lowest-income countries face significant increases in premature deaths, while some colder and higher-income countries see a slight reduction. This large disparity in premature deaths between high and low-income countries drives the increase in the SC-CO_2 under income-weighting. Weighting a dollar of damages in low-income countries the same as a dollar of damages in high-income countries across all four sectors yields a 2020 SC-CO_2 of \$218 per tCO_2, with 62% of damages coming from mortality. Income weighting increases the 2020 SC-CO_2 to \$527 when using global average income as the reference point and to \$3,224 when using U.S. income as the reference point, with 77% of damages coming from mortality. If the income-weighted SC-CO_2 values derived in this study were used in U.S. benefit-cost analysis, benefits would exceed costs for a much larger set of climate policies.

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1 Main

1.1 Introduction

A large body of literature finds that climate change will cause an increase in temperature-related mortality [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13], and that these deaths will be concentrated in low-income countries [2, 3, 6]. However, the three models used by the U.S. Government before 2023 to estimate the social cost of carbon (SC-CO_2)—the estimated monetized damage from emitting one tonne of carbon dioxide—either did not clearly specify how much, if any, of their estimated damages derive from mortality, or projected little damage from mortality [14].[1] In December 2023, the U.S. Environmental Protection Agency (EPA) published a report with updated methodology for estimating the SC-CO_2, increasing the central estimate from \$51 to \$190 per tCO_2 [15]. In this updated methodology, premature mortality is the largest source of damage. The SC-CO_2 is relevant to policy analysis because it can be used to monetize climate benefits in benefit cost analysis (BCA). In the United States alone, Federal regulations with benefits totaling well over \$1 trillion have incorporated the SC-CO_2 into Federal economic analysis, since the metric was first used [17].

Importantly, these SC-CO_2 estimates treat a dollar of damages in low-income countries the same as a dollar of damages in high-income countries. In contrast, income weights estimate a policy's impact on well-being by applying larger weights to damages on lower-income people than damages on higher-income people, which has a long tradition in the climate economics literature [18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29]. They have been adopted in some countries, such as the United Kingdom [30] and Germany [31], and are considered advisable by the Intergovernmental Panel on Climate Change [26].[2]

In this study, we combine three crucial elements: (1) global probabilistic socioeconomic projections and emissions projections [34], a climate model [35, 36], and sector-specific damage functions [37] all updated to the latest science and responsive to the National Academies of Sciences' recommendations for improving the scientific basis for estimating the SC-CO_2 [38]; (2) global mortality damage functions with country-level spatial resolution that explicitly account for cold- and heat-related mortality and income-based adaptation [2], and (3) income-weighted SC-CO_2 estimates. We find a large disparity in temperature-related deaths between

[1] Two of those models, DICE and PAGE, did not clearly specify how much (if any) of their damages came from mortality [1, 15], while the third, FUND, projected little damage from mortality [16].

[2] In November 2023, the U.S. White House Office of Management and Budget updated BCA guidelines across the Federal government, which sanctioned the use of income weights for the first time in the U.S. as either the primary or supplemental estimate of net benefits [32]. In January of 2025, however, this guidance was rescinded [33]. Irrespective of its use in the U.S., income weighting remains an important part of the literature and policy practice in multiple countries.

high-income and low-income countries, which drives the increase in the SC-CO_2 under income-weighting.[3]

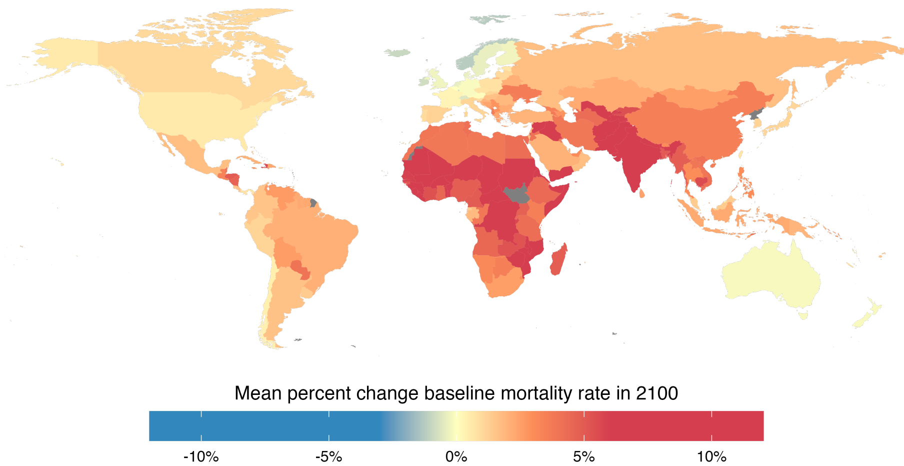
1.2 Results

Physical Mortality Projections Although climate change is expected to cause a significant increase in heat-related deaths, an expected decrease in cold-related deaths and reduced vulnerability to heat due to rising incomes counteracts this trend [1, 3, 44, 5, 6, 8, 7, 9, 45, 46, 47, 48, 10, 49, 50, 51, 52, 53, 11, 12, 54, 13]. Here, we project global temperature-related deaths at the country level accounting for both decreased cold-related mortality as well as income-based adaptation to heat (Fig. 1b). Rising incomes can reduce vulnerability to heat through multiple pathways, including by enabling increased adoption of air conditioning [55], increasing the ability to relocate labor hours away from occupations and times of day most exposed to the heat [56], and increasing the ability to live in locations less susceptible to the urban heat island effect [57]. Because higher-income populations are better able to protect themselves from heat, not accounting for expected income growth—that is, assuming that historical levels of adaptation are simply preserved in the future and more affluent populations will be just as vulnerable to heat as historically observed populations—risks over-estimating mortality impacts. To make our projections, we run Monte Carlo simulations that capture uncertainty in emissions, population, economic growth, the response of the climate system, and damages (see Methods section for details).

The projected temperature-related mortality impacts from climate change are unevenly distributed across the globe. The most severe mortality impacts are concentrated in hotter and lower-income countries. When accounting for income-based adaptation to heat (Fig. 1b), some colder and higher-income countries see a slight decrease in premature mortality: the benefit of fewer cold-related deaths outweighs the cost of more heat-related deaths. For instance, in 2100, we project 3,000 fewer premature deaths in Canada (a 0.7% decrease in the all-cause mortality rate) and 1,000 fewer premature deaths in Norway (a 1.6 % decrease). For lower-income and hotter countries, however, less exposure to cold temperatures does not overcome the burden of exposure to higher temperatures. Income-based adaptation

[3] While distributional equity has been explored in the SCC-CO_2 literature, previous studies have not implemented a mortality damage function like the one we employ here. The most closely-related paper is [39], which income-weights the SC-CO_2 but uses a mortality damage function with only nine region coefficients. Other studies examine distributional aspects of the SC-CO_2: [40] models how governments might choose domestic versus global carbon prices under different welfare objectives; [41] and [42] allocate non-income-weighted SCCs across countries; [43] shows how within-country inequality can amplify the optimal carbon price in a theoretical tax framework. This paper therefore advances this line of research and links it to the wider income-weighting literature that has long emphasized the ethical importance of declining marginal utility in climate-damage valuation (see, e.g., [27, 19].)

(a) Without income-based adaptation



(b) With income-based adaptation

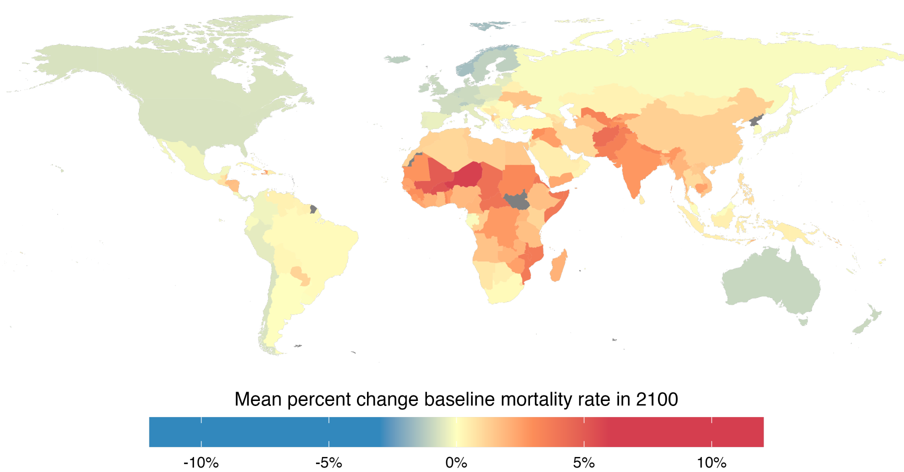


Fig. 1 | The spatial distribution of temperature-related mortality impacts in 2100. Maps show the mean estimated percent increase in the temperature-related mortality rate due to climate change in 2100. We take averages across the 10,000 draws in a Monte Carlo simulation, which captures uncertainty in socioeconomic and emissions scenarios (RFF-SPs), uncertainty in climate (FaIR v1.6.2), and damage function parameters (see Methods section for details). Countries not included in the model are shown in gray. **(a)** Shows results without accounting for income-based adaptation. **(b)** Shows results accounting for income-based adaptation.

plays a role in reducing this harm, but many countries still experience a significant increase in net mortality. We project the most harm in countries in Africa, the Middle East, and South Asia. Niger has the highest increase in mortality rate, an expected 5.9% increase (48,000 additional yearly premature deaths) in 2100. Countries with some of the largest populations in the world experience significant harm, including Pakistan (3.5% increase, 170,000 additional yearly premature deaths), India (2.8% increase, 544,000 additional yearly premature deaths), Nigeria (1.9% increase, 121,000 additional yearly premature deaths), and China (1.3% increase, 185,000 additional yearly premature deaths).

When aggregating premature deaths across all countries to project the net global mortality impact, we find that the increase in premature deaths in hotter and lower-income locations exceeds the decrease in premature deaths in colder and higher-income locations. Climate change causes a significant number of premature deaths even after accounting for future income-based adaptation (Fig. 2). Without accounting for the benefits of future income growth, and instead assuming that currently observed temperature-mortality relationships hold into the future, we project that climate change will cause 4.8 million premature deaths per year in 2100 (1.2M–10.4M: 5%-95% range) and 181 million total premature deaths from 2023–2100 [57M–363M 5%–95% range]. When we account for income-based adaptation, we project that climate change will cause 1.6 million temperature-related deaths per year in 2100 (-1.5M–5.6M: 5%–95% range) and 85 million total premature deaths from 2023–2100 (-20M–225M 5%–95% range). All results hereafter include income-based adaptation when projecting mortality impacts.

SC-CO_2 Results Because the SC-CO_2 is used to account for climate impacts in benefit-cost analysis (BCA), analysts that want to conduct an income-weighted BCA require income-weighted SC-CO_2 values to ensure that the costs and benefits of the action being analyzed—both climate and non-climate—are treated consistently. Income weighting involves two main steps (see Methods section for more detail) [27, 21, 24, 25, 18, 19, 28, 29, 22, 26, 20, 23, 39]. First, it accounts for diminishing marginal utility: it captures that an additional dollar has more value to someone who has a lower income than someone with a higher income. Income weighting uses a utility function to determine the effect of damages on well-being based on an individual's income. All else equal, the more that climate damages fall on low-income relative to high-income people, the higher the well-being loss. Second, income weighting converts these utility-denominated damages back into units of money, which can more easily be used in BCA. It does this by converting utility-denominated damages into units of money as valued by some person with some level of income (who is often represented as a person of average income in some "reference region", which is also sometimes referred to as a "normalization region")

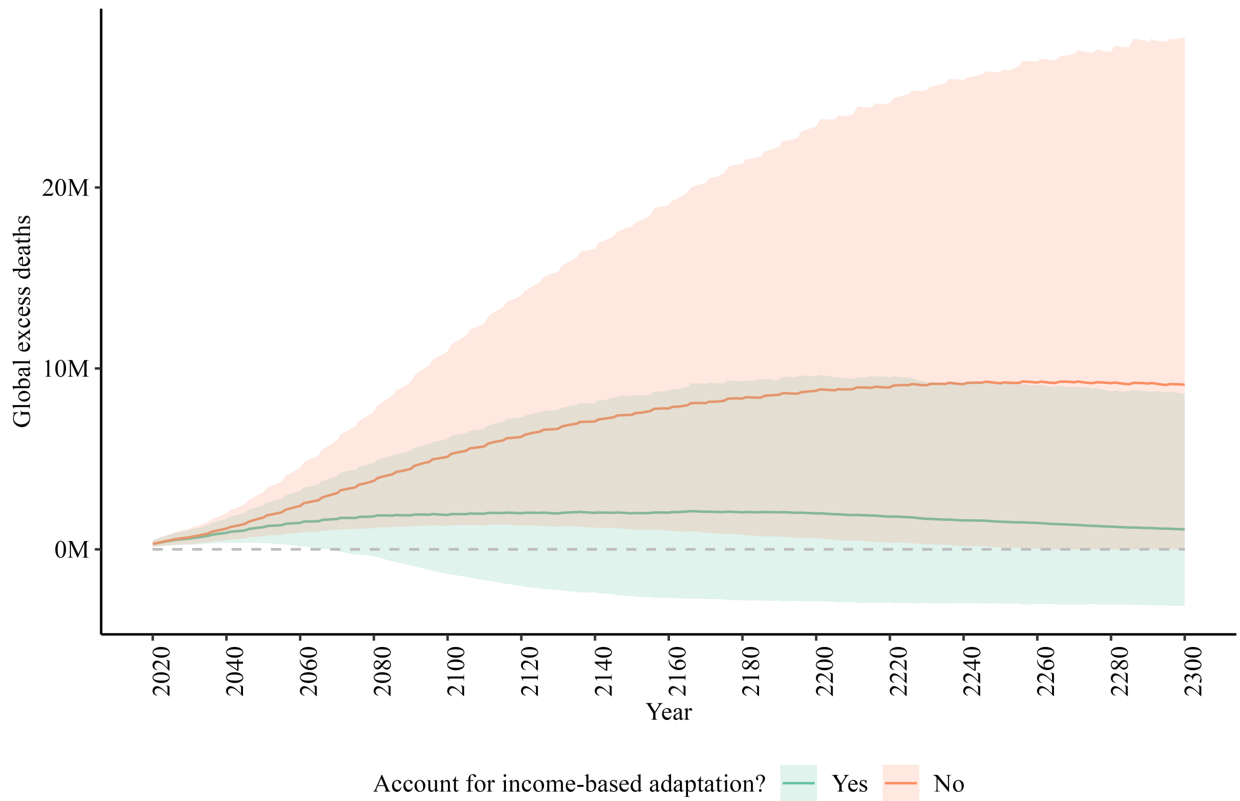


Fig. 2 | Yearly Global Premature Deaths From Climate Change, 2023–2300. Lines represent mean projections, and shaded regions represent the 5th-95th percentile projections. Projections that do not account for income-based adaptation are shown in red. Projections that account for income-based adaptation are shown in green.

[18, 20, 58, 23, 28, 59]. In this study, we refer to this level of income as the reference point. Using a higher-income reference point increases the SC-CO_2 by virtue of higher-income people valuing an incremental dollar less, whereas using a lower-income reference point decreases the SC-CO_2 by virtue of lower-income people valuing an incremental dollar more. The choice of reference point does not affect projected damages; rather, it changes the units in which these damages are represented. Different reference points do not alter the estimated damages caused by an additional tCO_2 the same way that expressing distances in miles or kilometers does not change the length of a trip [23].

Using the value of statistical life (VSL) to monetize premature mortality, we find that weighting a dollar of damages in low-income countries the same as a dollar of damages in high-income countries yields a 2020 SC-CO_2 of \$218 per tCO_2. When using global average income as the reference point, income weighting increases the SC-CO_2 by a factor of 2.4 to \$527. When using U.S. average income as the reference point, income weighting increases the SC-CO_2 by a factor of 14.8 to \$3,224 (Fig. 3). The highly unequal mortality effect of climate change is the primary driver of the increase in the SC-CO_2 when using income weights. Income weighting increases monetized temperature-related mortality damages more than any other sector: temperature-related mortality damages account for 88% of the increase in the income-weighted SC-CO_2 whereas all other damage categories (agriculture, energy, and sea-level rise) account for 12% of the increase. Temperature-related mortality makes up 62% of the SC-CO_2 before income weighting, but 77% of the SC-CO_2 after income weighting.

Table 1 holds all parts of the model fixed while varying the reference point. Income weighting using U.S. average income as the reference point increases the SC-CO_2 from \$218 to \$3,224 per tCO_2—a factor of 14.8—relative to the SC-CO_2 that values damages in low-income countries the same as a dollar of damages in high-income countries (what we refer to as "Equal Dollar Valuation"). Using Chinese average income as the reference point increases the SC-CO_2 by a factor of 2.2 (to \$471) and using Indian average income as the reference point decreases the SC-CO_2 by a factor of 0.7 (to \$160). In other words, the total social cost of emitting one tonne of CO_2 in terms of well-being is equivalent to causing \$3,224 in damages to an average American, \$471 to an average Chinese person, and \$160 to an average Indian person.[4] Table A.2 shows the income-weighted SC-CO_2 using all 184 countries in the model. One reason analysts may choose a particular reference point is based on which region will bear the cost of emissions reductions [18]. For instance, if emissions reductions are taking place in the U.S., using U.S. average income as the reference point converts this value into units of money as it is valued by a typical individual in the U.S. Therefore, under

[4] Note, these values are equivalency measures not desegregated impacts, and, therefore, should not be summed but considered within the context of choosing a reference point.

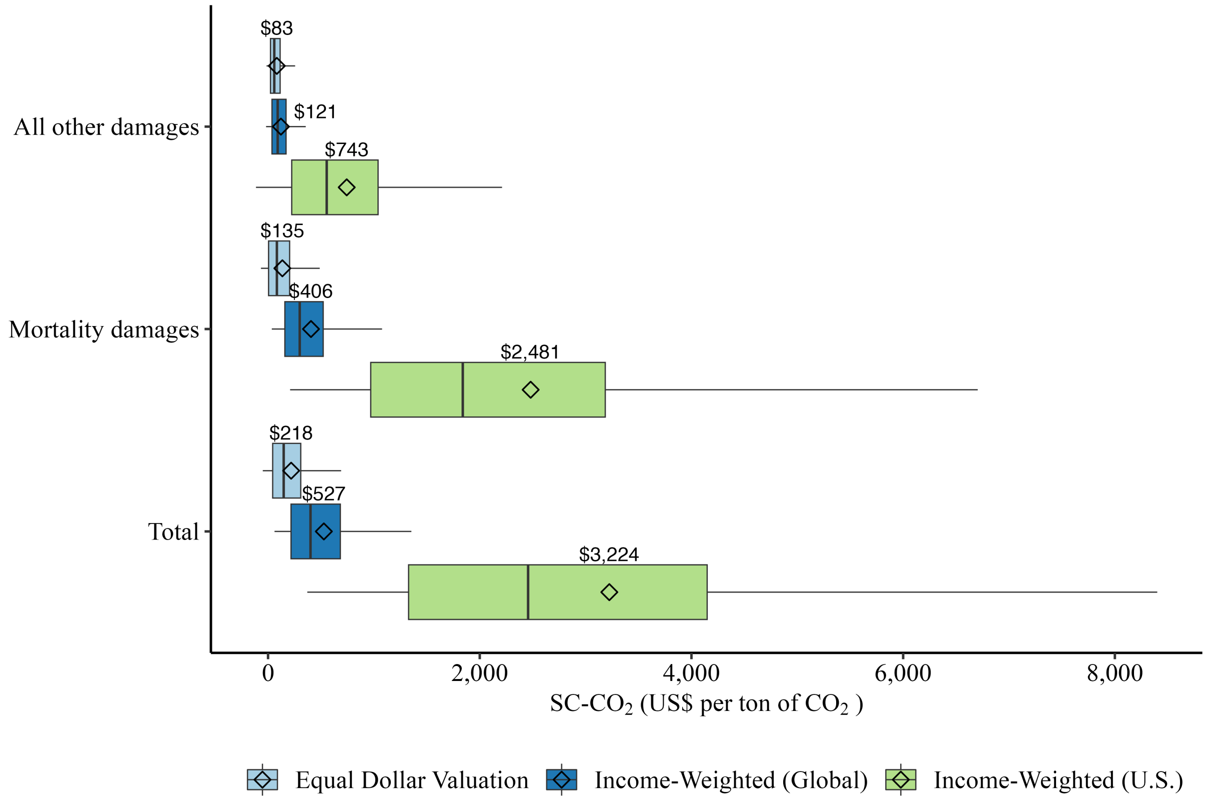


Fig. 3 | The 2020 SC-CO_2 by monetization choice. Boxplots show median (center black line), mean (black diamond), 25%–75% quantile range (box width), and 5%–95% quantile range (horizontal lines) SC-CO_2 values. The non-mortality values represent the partial SC-CO_2 aggregated across the three non-mortality sectors (agriculture, energy, and sea-level rise). The mortality values represent the mortality partial SC-CO_2. The total values represent the full SC-CO_2 across non-mortality and mortality. Light blue boxplots show equal dollar valuation SC-CO_2 values (i.e., counting a dollar of damages in poor countries the same as a dollar of damages in rich countries), the dark blue boxplots show income-weighted SC-CO_2 values using global mean income as the reference point, and green boxplots show income-weighted SC-CO_2 values with U.S. average income as the reference point. Values are expressed in 2020 US dollars per metric ton of CO_2. The figure uses a pure rate of time preference $\rho = 0.2\%$ (consistent with the main specification of [15] and [37]) and a utility curvature parameter of $\eta = 1.4$ (which reflects the OMB BCA guidelines from 2023 [32]). We show results across a number of alternative ρ and η values in the appendix, including ρ and η values used by the German government. Premature mortality is monetized using country-level estimates of VSL. Dollars are purchasing power parity (PPP) adjusted.

the framework presented in this paper, the cost of emissions reductions taking place in the U.S. do not need to be income-weighted (i.e., the income-weight for U.S. costs is unity) [18]. If, however, emissions reductions are taking place globally, and analysts wish to use a single global income-weighted SC-CO_2 value for BCA across countries, using global average income as the reference point converts this value into units of money as it is valued by the global average individual. Importantly, mitigation costs in the BCA also need to be income weighted using the same reference point. Analysts may also prefer to use an income-weighted SC-CO_2 with a global reference region if they prefer the efficiency of using a single SC-CO_2 regardless of where the emission reduction is taking place.

Table 1: The 2020 SC-CO_2 by reference region

Specification	SC-CO_2 ($ per tCO_2)
Equal Dollar Valuation	218
	[-50 - 690]
Income-Weighted (Global)	527
	[61 - 1,354]
Income-Weighted (U.S.)	3,224
	[371 - 8,401]
Income-Weighted (China)	471
	[53 - 1,243]
Income-Weighted (India)	160
	[18 - 419]

Table shows mean and 5th–95th quantile 2020 SC-CO_2 values across 10,000 Monte Carlo runs using a pure rate of time preference $\rho = 0.2\%$ and a utility curvature parameter of $\eta = 1.4$.

Fig. 4 holds all parts of the model fixed except the utility curvature parameter, η. A higher η indicates a more curved utility function. This implies that the marginal utility gained from a marginal dollar declines more rapidly compared to a lower η value. Before income weighting, η affects the SC-CO_2 in two ways [15, 37]. First, it accounts for diminishing marginal utility across time (see Methods section for details). A higher η tends to place less weight on future damages because a marginal dollar of damages to comparatively higher-income people in the future will be considered less harmful than a marginal dollar of damages to comparatively lower-income people today. Second, it accounts for risk aversion across uncertain potential future states of the world. A higher η implies higher risk aversion. Income weighting adds a third role for η: it accounts for diminishing marginal utility across space. That is, a higher η places more weight on damages in low-income locations [22, 26].

In Fig. 4, we vary η from 1 to 2, the range of commonly-used η values in BCA, holding the reference point constant at global average income (see Methods for details). Before

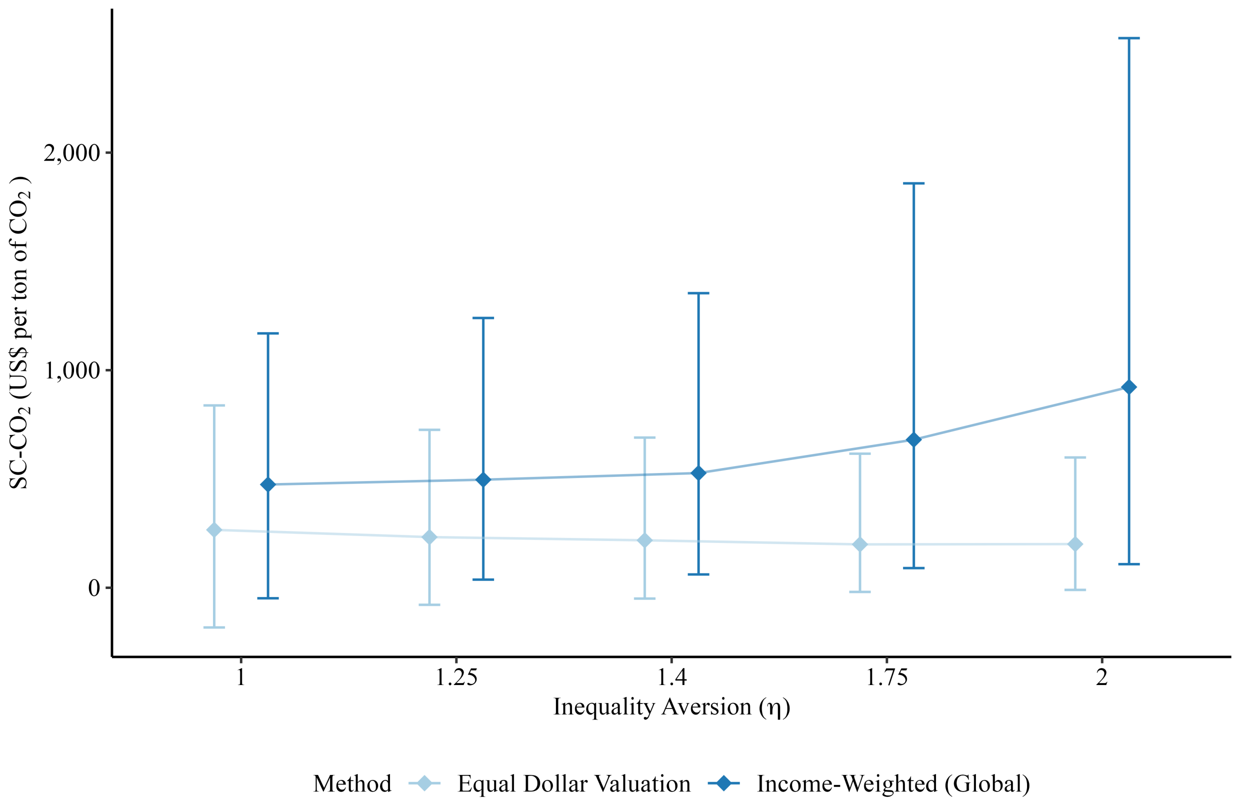


Fig. 4 | Estimates of the 2020 income-weighted SC-CO_2 vary with inequality aversion. The reference income level is held constant at global average income. The pure rate of time preference is held constant at $\rho = 0.2\%$. The diamonds represent mean SC-CO_2 values, and the error bars represent the 5%–95% quantile range.

income weighting, a higher η results in a lower SC-CO_2, as shown in Fig. 4. After income weighting, however, a larger η results in a higher SC-CO_2. This implies that η's additional role in putting more weight on damages in poor locations (interspatial inequality aversion), added by income weighting, is enough to outweigh η's impact on intertemporal inequality aversion that places less weight on future damages and risk aversion. Extended Data Table A.1 shows the estimates of the income-weighted SC-CO_2 using the η values from the EPA's 2023 report and currently in use in Germany, 1.24 and 1, respectively. A large literature highlights intergenerational inequality aversion as a key driver of the magnitude of the SC-CO_2 [60, 61, 62, 63]. Our findings suggest that choices around accounting for interspatial inequality aversion are similarly important in driving SC-CO_2 results.

1.3 Conclusion

We emphasize several important caveats to our findings. First, our mortality results only include premature deaths from temperature-related mortality. We do not include other potentially important climate-mortality pathways, including the effect of climate change on infectious disease, civil and interstate war, and food supply.

Second, following most of the epidemiology and economics literature [1, 3, 4, 5, 6, 7, 9], we make mortality projections based on dry-bulb temperature and not wet-bulb temperature. Wet-bulb temperature accounts for the role of atmospheric conditions, including humidity, in facilitating evaporative cooling. Although wet-bulb temperature has been identified as an important metric with regard to temperature-related mortality and in defining thresholds for uncompensable heat stress [64, 65, 66, 67, 68], few empirical studies on temperature-related mortality have assessed the impact of wet-bulb temperature [69, 64].

Third, the uncompensable heat stress thresholds identified in the applied physiology literature [68] suggest that the most extreme future temperatures may be far more damaging than the existing projections [65, 70], which are used to calibrate the mortality damage function in this study as well as the other studies in the SC-CO_2 literature [1, 3, 37] (see section 2.2.5 for a detailed discussion.) Accounting for infectious disease, conflict, food supply, wet-bulb temperature, and uncompensable heat stress may further increase projected mortality damages beyond what we estimate in this paper.

Fourth, our results account for income heterogeneity at the national level. Accounting for sub-national income heterogeneity would likely increase the income-weighted SC-CO_2 given climate damages—especially heat-related mortality—disproportionately burden the poorest populations within countries.

Finally, like the rest of the SC-CO_2 literature, our mortality damage function is derived from temperature-mortality exposure response functions that are estimated in the countries where the required high-frequency mortality microdata is available, which in our case are 23 countries that together represent around 40% of the world's population (see section 2.2 for details). We extrapolate temperature-mortality relationships from the parts of the world where data is available to the parts of the world where data is not available.

Overall, our findings suggest that, relative to counting a dollar of damages in low-income countries the same as a dollar of damages in high-income countries, income weighting substantially increases the SC-CO_2 when U.S. income, global average income, or the incomes of most countries in the world are used as a reference point (Table A.2). The disproportionate mortality effect of climate change accounts for 88% of the increase in the income-weighted SC-CO_2. If the U.S. were to use income-weighted SC-CO_2 values in benefit cost analysis consistent with the social welfare function adopted in this paper, the estimated benefits of greenhouse gas mitigation would substantially increase, causing benefits to exceed costs for a much larger set of climate regulatory, budgeting, and procurement policies. This could, in turn, have massive implications for the speed and magnitude of decarbonization in many high-income countries across the globe.

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2 Methods

2.1 Socioeconomic Projections and Climate Models

We use the Resources for the Future Socioeconomic Projections (RFF-SPs) as our source for probabilistic GHG emissions, economic growth, and population projections [37, 34]. The RFF-SPs are used in both the US EPA's 2023 SC-CO_2 report [15] as well as in the original version of the Greenhouse Gas Impact Valuation Estimator (GIVE) model [37]. The RFF-SP's median 21st-century emissions trajectory is most similar to Representative Concentration Pathway (RCP) 4.5. Economic projections in the RFF-SPs are represented in purchasing power parity (PPP) adjusted dollars, which adjusts market prices to account for the ability of money in different places to purchase fixed bundles of goods and services. See [15, 37, 34] for more details.

We represent the global climate system and carbon cycle dynamics with the Finite Amplitude Impulse Response (FAIR) version 1.6.2, consistent with [37], which accounts for climate system uncertainty. Because the mortality damage function used in this study requires country-level temperature projections, we apply a temperature pattern scaling methodology [71] to recover country-level temperatures from the FaIR global temperature outputs, discussed below. Lastly, we make probabilistic projections of regional changes in sea level using the Building blocks for Relevant Ice and Climate Knowledge (BRICK) model [72], also consistent with [37] and discussed below.

2.2 Damage Functions

2.2.1 How Integrated Assessment Models Estimate Damages from a *Pulse* of Emissions

Because this study's main goal is to quantify the income-weighted SC-CO_2, it falls into an important sub-category of the literature that determines the impact of adding a small *pulse* of emissions, the monetized value of which is the SC-CO_2. There are only a few studies that calculate the SC-CO_2 while including the mortality impacts of climate change updated to recent scientific understanding [1, 3, 37], and these are the closest studies to this study in the literature. Because SC-CO_2 models are focused on determining the impact of a pulse of emissions, they must make different design choices from other studies not focused on quantifying the impact of a pulse of emissions. Each of these studies, including this study, follows a three-step process.

Step 1 of the process is to gather decades of geographically resolved daily data on temperature and mortality from as many locations as possible, and then, using statistical methods (in our case distributed-lag nonlinear models) to establish relationships between daily exposure to hot and cold temperatures and premature mortality. Then, future changes in heat and cold-related mortality are estimated by combining these empirically identified exposure-response functions (which show the relationship between hot and cold days and premature mortality in different locations) combined with future daily temperature projections in specific emissions scenarios to determine the impact of future climate change on premature mortality.

Step 2 makes these projections interoperable with the integrated assessment model (IAM) that is designed to determine the monetized damage of adding an extra pulse of emissions, i.e., the SC-CO_2. This is where the *mortality damage function*—a tool that is used by every study in this literature—comes in. For the reasons discussed in Section 2.2.2, one cannot simply take all of the operations from step 1 and embed them within an SC-CO_2 IAM. The mortality damage function is a bridge between this empirical literature that uses this extensive daily historical vital statistics and weather data to make projections, and the more aggregated climatic and socioeconomic variables that are available in SC-CO_2 models.

Step 3 uses the IAM to calculate the SC-CO_2 by integrating stochastic socioeconomic projections with all of the study's sectoral damage functions, pulses the baseline climate projection with a pulse of CO_2, and determines the impact on damages by discounting future damages. Step 1 is usually not performed by the integrated assessment modeling group itself; it is usually done independently of the work of the integrated assessment modeling group by other research groups that are subject matter experts in the specific damages category. Steps

2 and 3 are also often done in separate studies.[5] We will discuss our case in more detail in section 2.2.3 below. In the context of the temperature-related mortality damages estimated in this paper, step 1 was carried out by the Multi-City Multi-Country Collaborative research group in Gasparrini et al. 2017 [6]. Step 2 was done by some members of this research team in Bressler et al. 2021 [2]. Our study represents step 3, the final and most policy-important step, by creating the IAM that brings together these temperature-related mortality damage functions and embeds them in an integrated assessment model that calculates the SC-CO_2 across different approaches to valuing lives and livelihoods around the world.

2.2.2 The Mortality Damage Function is the Bridge that Links Daily Temperature Exposure-Response Relationships to SC-CO_2 Models

There is now a large empirical literature that uses Distributed Lag Non-Linear Models (DLNMs) and similar methods to estimate exposure-response relationships between daily temperature and mortality using historical data. Some studies combine their historically estimated exposure-response relationships with future temperature projections to estimate the total future impact of climate change on temperature-related mortality in specific scenarios, e.g., [5, 6, 7, 8, 9, 10, 11, 12, 77, 13] among others. While these studies use exposure-response functions *directly* to estimate the total temperature-related mortality impact of climate change in future emissions scenarios, all studies that estimate the marginal impact from adding a pulse of emissions (i.e., the SC-CO_2)[6] must use exposure-response functions *indirectly* through the use of a *mortality damage function* [1, 3, 37]. In each of these studies, future temperature-related mortality projections in specific future scenarios are used to calibrate a damage function. In the following paragraph, we discuss why all of the SC-CO_2 models that include mortality impacts [1, 3, 37] use mortality damage functions; we return to discuss how this calibration is done in our context in Section 2.2.3.

The mortality damage function serves as a bridge between the empirical literature and SC-CO_2 models by estimating climate-mortality impacts as a function of the climatic and socioeconomic variables available in SC-CO_2 models. For instance, all of the latest generation SC-CO_2 models [3, 15, 37] use the FAIR climate model [35] to estimate the climatic impact of a pulse of emissions. The FAIR model is a reduced-complexity climate model that determines

[5]For example, for step 2, [37] relied on sectoral damage functions that were published in multiple previous studies [73, 4, 74, 75] and the [37] study itself represented just step 3. Whereas for [1], the economic damages portion of step 2 was done in a previous study [76] while the mortality damages portion of step 2 was done in that study itself along with step 3.

[6]And include temperature-related mortality impacts as part of their SC-CO_2 estimates. Until recently, most SC-CO_2 models largely ignored temperature-related mortality impacts. See section 1.1 for more discussion on this point.

the impact of adding a pulse of emissions at some point in time on *global mean surface temperature* in every year out to 2300. The FAIR model represented a significant improvement over the models used in previous generation integrated assessment models (see [78, 79] for details), and was specifically recommended by the U.S. National Academies of Sciences for use in calculating the SC-CO_2 [38]. However, if one were to use empirically estimated exposure-response functions instead of a mortality damage function directly in an SC-CO_2 model, they would need much more than the impact of a pulse of emissions on global mean surface temperature. They would need not just the projections of daily temperatures in every country out to 2300, but also the impact that adding that pulse of emissions would have on daily temperatures in each country to the end of the model period in 2300. Indeed, there are climate models that can estimate daily temperatures out to 2100 in specific emissions scenarios, e.g., the ISI-MIP projections used by [6] and the Nex-GDDP projections used by [77], but these models do not capture the impact of adding an extra *pulse* of emissions on daily temperatures to the end of the model period, which is required to calculate the SC-CO_2.

Still, our study innovates on the existing literature by downscaling the climatic impacts of a pulse of emissions from global mean surface temperature to the country level (discussed in detail in Section 2.2.3).[7] However, neither our model nor any other model in the SC-CO_2 literature produces the impact of a pulse of emissions on daily temperatures, which would be required if one were to use exposure-response functions directly to estimate mortality impacts instead of a mortality damage function.

Furthermore, even if one had a climate model that could quantify the impact of a pulse of emissions on daily temperatures, they would need to carefully think through trade-offs between adding complexity to the model versus accounting for climatic, socioeconomic, and damages uncertainty while also producing SC-CO_2 estimates within a reasonable runtime period[8], which is important both for reproducibility and the wider usage of the models and their results. Indeed, the National Academies of Sciences Committee on "Assessing Approaches to Updating the Social Cost of Carbon" emphasized that the SC-CO_2 models that were being used by the U.S. government until 2023 were not adequately capturing uncertainty [38]. All of the newest generation SC-CO_2 models that have been developed within the last few years and were used by the U.S. EPA in the 2023 report made significant improvements

[7] While all of the latest-generation SC-CO_2 models use only global mean surface temperature to determine the impact from a pulse of emissions, older-generation Fund 3.8 SC-CO_2 model that the U.S. government used until 2023 [80] downscaled global mean surface temperature to the level of 16 regions by using a simple constant derived from 14 GCMs in 2000 [81]. Our downscaling procedure described in section 2.3 represents an improvement and modernization from this past literature.

[8] Even with the FAIR model, each of the many SC-CO_2 estimates provided in this study (i.e. the estimate of a specific SC-CO_2 in a specific year) take at least 12 hours of runtime, which is similar to, e.g., Rennert et al. 2022 [37].

over previous generation models in capturing uncertainty by running 10,000 Monte Carlo simulations sampling across climatic, socioeconomic, and damage uncertainties.

2.2.3 This Study's Mortality Damage Function

We leverage the Bressler et al. 2021 [2] mortality damage function in this study. We will discuss it briefly here and point the reader to the study for further details. Bressler et al. 2021 produced temperature-related mortality damage functions at the country level, which estimate the impact of climate change on both increasing heat-related mortality and decreasing cold-related mortality. Furthermore, they make projections that account for income-based adaptation in different parts of the world, i.e., the benefit of future income growth in reducing vulnerability to heat-related mortality. They can also make projections assuming that the current vulnerability to temperature-related mortality remains the same in the future, i.e., that no additional adaptation from future income growth will occur.

This mortality damage function was calibrated based on the projections from Gasparrini et al. 2017 [6]. Gasparrini et al. 2017 make those projections by using their large dataset of high-frequency and geographically resolved historical vital statistics and weather data across their 23 countries to estimate exposure response functions by using DLNM models to estimate the relationship between hot and cold temperatures and premature mortality. They then project future excess mortality by using the exposure response functions that were estimated from historical data, but applying the projected future distribution of temperatures from climate model outputs in specific emissions scenarios.

As mentioned above, it is not feasible to directly use exposure-response functions in the context of calculating the SC-CO_2. Thus, Bressler et al. 2021 focused on accurately calibrating the hot and cold-related mortality damage functions to the results from Gasparrini et al. 2017, and the validation exercises in that study involve making projections of heat and cold-related mortality in specific scenarios and then comparing that prediction versus what Gasparrini et al. 2017 predict in that same scenario. Figure 1 in Bressler et al. 2021 shows the performance of the preferred heat and cold damage function models versus the projections made in Gasparrini et al. 2017.

Because the focus of the Bressler et al. 2021 mortality damage function was prediction, the main approach that they used when analyzing model performance and determining model selection was cross-validation, which involves splitting the data into training data that is used to calibrate the model, and then the model is used to make predictions on the data that is not used to train the model (model performance on cross-validation is shown in Bressler et al. 2021 tables 1 and 2). For example, if we were to drop Thailand, re-fit the model on the remaining 22 countries, and then predict Thailand's temperature-mortality response, the typical error is 2 percentage points for heat and <0.5 percentage points for cold. These tests give us a realistic sense of performance when extrapolating to places with similar climates/incomes that were not in the estimation sample.

Additional model performance considerations in Bressler et al. 2021 included aggregate plausibility and sensitivity analyses. They assessed aggregate plausibility in comparison to the peer literature by rolling up the country-level projections up to the globe, and found that the end-of-century net increase in mortality (4% under RCP 8.5) is close to two other estimates that use different methods and data [1, 3]. For sensitivity analyses, Tables 1 & 2 in Bressler et al. 2021 showed alternative model specifications, all of which performed worse out-of-sample.

In this study, we use the preferred models from Bressler et al. 2021 for both heat (model 4) and cold (model 3). Formally, the model for heat-related mortality is:

$$\begin{aligned} Y_{c,t}^{\text{Heat}} &= \beta_1 T_{c,t} + \beta_2 T_{c,t}^2 + \beta_3(\text{Hottest Month Avg Temp}_c) \\ &\quad \beta_4 T_{c,t}(\text{Hottest Month Avg Temp}_c)(\log(y_{c,t})) \end{aligned} \tag{1}$$

where $Y_{c,t}^{\text{Heat}}$ is the percentage increase in the all-cause mortality rate due to heat in country c at time t, $T_{c,t}$ is the increase in yearly average temperature relative to the 2001-2020 period in country c at time t, Hottest Month Avg Temp$_c$ is the population-weighted average temperature in the hottest month in country c between 1984 and 2015, and $y_{c,t}$ is the PPP adjusted per capita GDP.[9] See [2] for further details. For our projection results that do not account for additional income-based adaptation, we hold $y_{c,t}$ constant at current levels. For our projection results that do account for income-based adaptation, we use the model's projections of future country-level income in each year for $y_{c,t}$.

The model for cold-related mortality is:

$$Y_{c,t}^{\text{Cold}} = \beta_1 T_{c,t} + \beta_2 T_{c,t}^2 + \beta_3(\text{Coldest Month Avg Temp}_c) \tag{2}$$

To represent damage function uncertainty in our Monte Carlo runs, a vector of coefficients for the heat and cold models is sampled from a multivariate normal distribution centered on the point estimate and standard deviation equal to the reported standard error in [2]. The net percentage increase in mortality rate, $Y_{c,t}$, is the sum of $Y_{c,t}^{\text{Heat}}$ and $Y_{c,t}^{\text{Cold}}$. After

[9] Population-weighted average temperatures are not available for seven countries, although population-weighted average temperatures are needed to calculate the hottest and coldest month for each country. We assign these seven countries the values of their neighbor as follows: Aruba is assigned Venezuela, Bahrain is assigned Saudi Arabia, Barbados and St. Lucia are assigned Puerto Rico, the Maldives are assigned Sri Lanka, Malta is assigned Tunisia, Singapore is assigned Malaysia, and Tonga is assigned Fiji.

the calculation of $Y_{c,t}$, the net percentage increase in mortality rate is converted into net additional deaths:

$$\text{Excess deaths}_{c,t} = (\text{Population}_{c,t}) \times (\text{Baseline mortality rate}_{c,t}) \times \left(\frac{Y_{c,t}}{100}\right) \tag{3}$$

where baseline mortality is defined as the country population level times its baseline mortality rate from the RFF-SPs. Since we run 10,000 Monte Carlo simulations, which capture uncertainty in emissions, population, economic growth, the response of the climate system, and damage functions, we estimate equation 3 for each country in each time period 10,000 times. Fig. 1 shows probabilistic globally aggregated projections across time.

In Bressler et al. 2021, damage functions were tested in making temperature-related mortality projections out to 2100 using deterministic SSP-RCP socioeconomic projections [2]. In that study, heat-related mortality was always increasing and cold-related mortality was always decreasing under increasing temperatures, which is consistent with Gasparrini et al. 2017 [6]. This study, however, makes projections out to 2300 using stochastic socioeconomic projections across 10,000 Monte Carlo draws. Because equations 1 and 2 are nonlinear, we seek to ensure that in the most extreme Monte Carlo runs, the model does not produce theoretically inconsistent behavior simply due to the nonlinear functional form. Thus, we impose an additional functional form restriction that ensures that holding all else equal, including income, heat-related mortality does not rise under higher temperatures (i.e., ensuring that the partial derivative of heat-related mortality with respect to temperature does not decrease under higher temperatures) and cold-related mortality does not fall under higher temperatures (i.e., ensuring that the partial derivative of cold-related mortality with respect to temperature decreases under higher temperatures). We find that imposing this restriction only has a small impact on results: it raises the 2020 SC-CO_2 by 2.8%.

Like much of the epidemiology and economics literature (e.g., [5, 6, 7, 9]) and all of the latest SC-CO_2 models that include mortality impacts [1, 3, 37], we make mortality projections based on dry-bulb temperature and not wet-bulb temperature. This is for multiple reasons. First, SC-CO_2 integrated assessment model climate modules that determine the temperature response to a pulse of emissions make dry-bulb temperature projections, as opposed to wet-bulb temperature projections, and wet-bulb temperature projections, would be necessary to operationalize mortality damage functions that use wet-bulb temperature. Second, very few empirical studies on temperature-related mortality have assessed the impact of wet-bulb temperature [69, 64], although wet-bulb temperature has been identified as an important metric with regard to temperature-related mortality and uncompensable heat stress [64, 65, 66, 67, 68].

Following the Gasparrini et al. 2017 study that is the precursor to the Bressler et al. 2021 mortality damage function that we leverage in our study, we assume that the historical relationship between population vulnerability and climate remains constant into the future. We do not account for population aging in our projections, and we leave this to future work. The literature that assesses the age-specific mortality burden from climate change is quite active and continuing to develop. Multiple past studies have found that elderly individuals are the most vulnerable to the most extreme heat exposure [3, 8, 45, 48, 10, 82, 83]. However, recent work in Mexico [77] found that while the elderly were the most vulnerable to the most extreme heat as prior work has shown, younger individuals were far more vulnerable to very hot days that were not the most extreme, but which occurred far more frequently. Because of this, they found that the overall mortality burden from heat was far higher among younger individuals than older individuals, and that future warming in Mexico is expected to shift the temperature-related mortality burden away from older individuals (who are comparatively more vulnerable to cold-related mortality) to younger individuals (who are comparatively more vulnerable to heat-related mortality). Subsequent work found that exposure to heat through physical labor is a major driver of this result [84]. We expect there to be continued work on projections of the age-specific mortality burden from climate change in other countries using granular age categories, and as the coverage of such work expands globally, we will return to add age-specific mortality damage functions in our model in future work.

2.2.4 Relation to Other Mortality Damage Functions in the SC-CO_2 Literature

Before this study, mortality damage functions in the latest generation SC-CO_2 models were functions of global mean surface temperature. For example, the mortality damage functions in Bressler 2021[10] [1] and Carleton et al. 2022 [3] are quadratic functions of global mean surface temperature.[11] Rennert et al. 2022 [37], which uses the mortality damage function derived in Cromar et al. 2022 [4], is a linear function of global mean surface temperature. This study advances the literature by being the first to estimate the mortality damage function as a function of country-level income as well as country-level climatic variables. This is not to say that Bressler 2021, Carleton et al. 2022, and Rennert et al. 2022 assume that yearly global mean surface temperature is the only relevant metric with respect to temperature-related mortality, nor that mortality damages are assumed to be the same in every other part of the world. To the contrary, those studies relied on empirically-grounded exposure-response

[10]Note that Bressler 2021 "The mortality cost of carbon" [1] is a separate study from Bressler et al. 2021 "Estimates of country level temperature-related mortality damage functions" [2], as we refer to both studies in this section.

[11]See Bressler 2021 [1] figure 6 and Carleton et al. 2022 [3] equation 9.

functions that were estimated based on historical observations to make projections into the future in a variety of regions.[12]

Constructing a mortality damage function as a function of global mean surface temperature is entirely valid, but it comes with some tradeoffs. To begin, an important advantage of making our mortality damage function a function of country-level income and country-level temperature is that we can dynamically estimate the mortality impact from a pulse of emissions while accounting for the specific income draw made in each of our 10,000 Monte Carlo simulations, [14] Whereas a mortality damage function that is only a function of global mean surface temperature is unable to connect the specific socioeconomic draw in each of the 10,000 Monte Carlo scenarios to its damages. For example, the Rennert et al. 2022 mortality damage function is only a linear function of global mean surface temperature, so it projects the same mortality damages regardless of future income levels, and because of the linearity assumption, it assumes that the mortality impact of slightly higher temperatures is the same regardless of how optimistic or pessimistic the future warming scenario is. The Carleton et al. 2022 mortality damage function, which was written before the RFF-SPs were available, is conditional on deterministic shared-socioeconomic pathway income projections. And the Bressler 2021 mortality damage function is a quadratic function of global mean surface temperature, so it also projects the same mortality damages regardless of future income levels. But in any case, it was integrated into the deterministic DICE model, which does not account for future uncertainty in temperatures or income.

In addition, our mortality damage function represents damages in *physical units* at the country level, i.e., it estimates the impact of a pulse of emissions on the mortality rate in each country. This allows us to easily and flexibly change the parameters that determine how the model values lives and livelihoods—the main aim of our study—without changing

[12]For example, Carleton et al. 2022 use exposure response functions estimated using data from countries that together cover nearly 40% of the world's population. They use relationships between local climate, income levels, and future climate projections to estimate mortality impacts in locations where they do not have data. However, in contrast to our study, they then monetize and aggregate their projected mortality impacts across the world to calibrate a single global mortality damage function, which represents the monetized mortality impact of climate change as a quadratic function of global mean surface temperature,[13] and it is this damage function that is used to estimate their partial mortality SC-CO_2. Likewise, Bressler 2021 [1] aggregates global mortality projections chosen from a systematic research synthesis of the climate mortality literature to calibrate its mortality damage function, which also estimates the global mortality impact of climate change as a simple quadratic function of global mean surface temperatures. The Rennert et al. 2022 [37] mortality damage function derived in Cromar et al. 2022 [4] breaks the world into nine regions, and uses regional and city-level empirical studies to calibrate nine mortality damage functions in each region, which are each a linear function of global mean surface temperature in the Rennert et al. 2022 implementation.

[14]As discussed in section 2.1, we use the probabilistic RFF-SP socioeconomic scenarios in our study, which were responsive to the National Academies of Sciences suggestions and which were used in the 2023 EPA report on the the SC-CO_2. These scenarios account for uncertainty in future economic outcomes by projecting income levels in each country over the model period in each one of the 10,000 Monte Carlo simulations.

other parts of the model. E.g., we can change a single parameter in our model to determine whether the SC-CO_2 is estimated with income weighting or weighting a dollar of damages in low-income countries the same as a dollar of damages in high-income countries. And we can change a single parameter to calculate the SCC across a variety of other parameters, such as the utility curvature parameter (η) and the time preference parameter (ρ), and we provide those sensitivities in the study. Whereas mortality damage functions that calculate *monetized* mortality damages have already presupposed their approach to valuing lives and livelihoods, and taking a different approach to valuing lives and livelihoods requires re-estimating the mortality damage function.

One additional distinction that it is important to draw in the temperature-related mortality literature—which is relevant for all of the mortality damage functions in the SC-CO_2 literature—is the difference between mortality *risk* to a given temperature exposure, and the overall projected mortality *burden* under climate change [77]. There are two crucial elements that determine the projected mortality burden from climate change, whether that is measured as a total mortality burden from a certain future scenario or whether that is measured as the mortality burden associated with adding a pulse of emissions, as in the case of the SC-CO_2: (1) The mortality risk from exposure to certain temperatures, which can be presented as relative risk as in [6, 85] or as absolute risk as in [3], and (2) the projected future exposure. Multiple studies have shown that hotter regions tend to exhibit less risk to a given high temperature exposure [3, 86, 87]. However, because those places are expected to be exposed to fewer of these high temperatures both now and in the future under climate change, the overall mortality burden is projected to be much higher in hotter places, although those places may tend to have less relative and absolute risk to a given day of heat exposure [3, 88]. Indeed, we also find that hotter and poorer regions are projected to have a higher mortality burden from climate change, consistent with the Gasparrini et al. [6] used to calibrate our mortality damage function.

Finally, one important caveat for all of the studies in this literature, including our study, is that because large parts of the world do not have the required data available to directly estimate temperature-mortality relationships (especially in the developing world), modelers must extrapolate temperature-mortality relationships identified in parts of the world where data is available to parts of the world where data is not available. In our case, the Gasparrini et al. 2017 study [6] was used to calibrate our mortality damage function [2] uses data from 23 countries to make projections in those 23 countries. These countries were selected based on the availability of daily mortality data and together make up around 40% of the world's population. High income countries such as those in Europe are well-represented, as are some low- and middle-income countries, including Thailand, The Philippines, Brazil, China, Mexico, and Vietnam.. However, no African countries are included. Indeed, if those countries did have this data available, then they would be used directly to estimate the mortality damage function. This further underscores the importance of the cross-validation exercise in Bressler et al. 2021 discussed in section 2.2.3.

2.2.5 Relation to the Uncompensable Heat Threshold Literature

An important strand of the temperature-related mortality literature that goes back at least to Sherwood and Huber 2010 [67] discusses the theoretical 35°C wet bulb temperature threshold, above which humans are no longer able to dissipate heat into the environment, and are thus physically incapable of survival when exposed for a sufficient length of time. This has led to important subsequent work.[15] This strand of the literature, which is focused on determining a threshold for uncompensable heat stress from physiological first principles, has developed separately from the much larger literature in epidemiology and economics that we use in this study to calibrate our mortality damage function.

There are multiple reasons why our study, as well as the rest of the related studies in the SC-CO_2 literature [1, 3, 37], use a mortality damage function derived from the empirical epidemiology and economics literature as opposed to the threshold-focused literature. First, the empirical literature has repeatedly shown that while the hottest and coldest days have the largest *marginal* impact on premature mortality, because those days occur so infrequently, the *total* mortality burden associated with the most extreme days is actually much smaller than the total mortality burden associated with quite hot/cold but not extreme hot/cold days that are less damaging on the margin, but which occur with much more frequency [77, 85]. For this reason, if one were to only consider extreme hot or extreme cold days, one may leave out the majority of temperature-related mortality damages. The mortality damage function here, as discussed in Section 2.2.3, is a mapping from specific warming/socioeconomic scenarios in each country and heat and cold-related mortality results derived from exposure response functions that account for the full distribution of cold and hot temperatures and the mortality impacts of exposure to each of these temperatures.

Furthermore, it is unclear how one would assign a level of damages to future mortality based on mortality projections that come from a threshold-based approach. One advantage of

[15] For example, Raymond et al. 2020 [66] showed that this theoretical threshold had been nearly reached historically in certain often sparsely-populated regions of the world for short periods of time. When applied physiologists have tested this theoretical threshold empirically on human subjects in labs, the wet-bulb temperature threshold for uncompensable heat stress has been found to actually be several degrees lower compared to the threshold theorized from physiological first-principles [68, 70]. Recent modeling studies, e.g., [65], used the threshold from the applied physiology literature combined with future climate simulations to determine when certain parts of the world will exceed this threshold in future climatic scenarios.

the exposure-response-function approach dominant in epidemiology and economics is that it makes a prediction of the number of deaths that occur in specific climatic scenarios based on extrapolating historically observed empirical relationships. Studies like [65, 67] consider when uncompensable heat-stress thresholds are projected to be breached in the future, but they do not project the number of deaths projected to occur when those thresholds are exceeded; this question is even more important in the context of calculating the SC-CO_2 because the mortality portion of the SC-CO_2 is concerned with estimating the marginal extra mortality associated with adding a marginal extra pulse of emissions, as opposed to determining the total mortality impact resulting from some future global emissions scenario. Previous work attempted to answer these questions and to reconcile the uncompensable-heat-threshold literature and the empirical exposure response function literature [77] by empirically assessing the mortality response to very high wet-bulb temperatures in Mexico, which has experienced some of the highest historical wet-bulb temperature exposure in the world [66]. Unfortunately, even in Mexico, extreme wet-bulb temperatures approaching uncompensable heat thresholds occurred so rarely in populated areas that there was not enough statistical power to address these questions [77].

However, is is crucial to emphasize that these uncompensable heat-thresholds represent an important reason why our temperature-related mortality estimates, as well as the temperature-related mortality estimates in the rest of the SC-CO_2 literature [1, 3, 37] may be underestimates. The empirical literature that these studies use to calibrate mortality damage functions relies on flexible functional forms, such as splines (in our case drawing from [6]) or higher-order polynomials (in the case of, e.g., [3]) to extrapolate exposure-response functions to temperatures outside of the support of historically observed temperatures. However, these uncompensable heat thresholds suggest that the most extreme temperatures in the future may be far more damaging than what the empirical epidemiology and economics literature project [65], in particular in the hotter and poorer countries where these thresholds are projected to be exceeded first [70]. The reconciliation of the empirical exposure-response function-based approach and the uncompensable-heat-stress-threshold-based approach may not be fully possible unless such uncompensable heat thresholds occur for sustained periods in sufficiently populated locations.

2.2.6 Monetizing Premature Deaths

Excess deaths are then monetized using the value of statistical life (VSL). We use the same approach to monetizing premature deaths used by EPA's 2023 SC-CO_2 report, which is the same method used in other major studies in this literature, such as Rennert et al. 2022 [37]. Our goal is to show how income-weighting changes the SC-CO_2 from the figures that are widely used in policy and in the literature while holding as much of the other choices fixed as possible. The baseline VSL for 2020 for the U.S. is derived using EPA's 1990 Guidance value of \$4.8 million and adjusted for income growth and inflation, resulting in a 2020 U.S. VSL of \$10.05 million [15]. We then adjusted for country c's GDP per capita in year t as

$$\mathrm{VSL}_{c,t} = \mathrm{VSL}^{\mathrm{base}}_{US,2020} \times \left(\frac{y_{c,t}}{y_{US,2020}}\right)^{\epsilon}. \tag{4}$$

In Table 2, we run a sensitivity analysis leveraging the approaches recommended by Robinson et al. 2019 [89] to show the sensitivity of our results to alternative VSL specifications. The main VSL specification that we use for all of the SC-CO_2 estimates in the main body of our paper (and that EPA 2023 and Rennert et al. 2022 use) is consistent with Robinson et al.'s option (b). The below table shows how the income-weighted SC-CO_2 varies when we hold all other aspects of the model the same (using U.S. average income as the reference region), but use alternative VSL approaches.

Table 2: The 2025 SC-CO_2 with alternative VSL assumptions

Specification	**SC-CO_2 (\$ per tCO_2)**
$\mathrm{VSL}^{\mathrm{base}}_{\mathrm{US},2020} = \$10.05$ million, $\epsilon = 1$	3,567
$\mathrm{VSL}^{\mathrm{base}}_{\mathrm{US},2020} = \$10.05$ million, $\epsilon = 1.5$	1,957
$\mathrm{VSL}^{\mathrm{base}}_{\mathrm{US},2020} = \$6.28$ million, $\epsilon = 1$	2,545

Table shows mean 2025 SC-CO_2 values across 10,000 Monte Carlo runs using a pure rate of time preference $\rho = 0.2\%$ and a utility curvature parameter of $\eta = 1.4$ with alternative VSL assumptions.

2.2.7 Seal Level Rise, Energy, and Agricultural Damage Functions

The sea-level rise, energy expenditures, and agricultural damage functions used in this study are discussed in Rennert et al. 2022 [37]. See Rennert et al. 2022 Extended Data Table 2 for a discussion of how uncertainty is represented in those damage functions.

2.3 Temperature Pattern Scaling

The temperature-related mortality damage function used in this study [2] requires country-level mean surface temperature ($CMST$) as an input, but the underlying reduced-complexity climate model(FaIR) produces projections of global mean surface temperature ($GMST$). To downscale $GMST$ to $CMST$, we incorporate the spatial patterns from high-resolution, spatially resolved surface temperatures from the sixth phase of the Coupled Model Intercomparison Project (CMIP6) general circulation and earth system models [90]. These models project climate futures at fine spatial and temporal resolutions but are large and computationally expensive and, as such, cannot be directly integrated into a climate-economy models like ours nor used in a probabilistic setting as we do in this study. One long-standing solution to this problem is to use a pattern scaling approach. This method is an efficient and effective way to emulate annual mean climate variables at local resolutions using outputs from reduced-complexity climate models that only output climate variables at the global level [91, 92, 93, 71]. We adopt the methods and code of [93] to recover local temperature patterns for the suite of CMIP5 models, and simply substitute the CMIP5 library with the updated CMIP6 library.

For each of the 21 available CMIP6 models, we regress the local mean surface temperature $LMST$ in grid cell i of each GCM/ESM model output in year t on the corresponding $GMST$:

$$LMST_{it} = \alpha_{it} + \sum_i \beta_i GMST_t[gridcell = i] + \varepsilon_{it}. \tag{5}$$

This results in a time-invariant vector of β's that dictate the relationship between $GMST$ and $LMST$ in each grid cell i, with α_{it} capturing the model's intercept and ε_{it} being the remaining variation in $LMST$ not explained by the regression. Because the spatial resolution of each ESM/GCM varies by model, we then overlay the model-specific spatial grid of β_i's with a uniform 1-degree by 1-degree grid and recover the average β_i's within each of the 1 degree cells—indexed below with k. This step is necessary to create a spatially-consistent layer of β's to use in the next step of the weighted aggregation.

To match the level of the β's with the level of the temperature-related mortality damage function, we must calculate an average of the β_k's across all grid cells within each country. Because temperature-related mortality impacts affect people and not land, we weight the aggregation of the β_k's using spatially-explicit, sub-national population centers observed in the year 2000 [94] and recover a vector of population-weighted, country-level β's, indexed by c below.

The steps taken above result in 21 unique country-level patterns, one for each CMIP6 model, that are population weighted averages to reflect the local temperatures faced by residents of each country. From the available set of 21 patterns, we randomly select one pattern to use in each Monte Carlo simulation of the model, indexed by s, providing another source of climate module uncertainty. We recover $CMST$ in each simulation s for each county c in model year t by scaling $GMST$ with the population-weighted country-level β's such that

$$CMST_{sct} = \beta_{sc} \times GMST_{st}. \tag{6}$$

The $CMST_{sct}$'s described here correspond to the T's in equations 1 and 2. The set of 21 CMIP6 models used reflect the entire set of available CMIP6 models available at the time the study was performed. That is, all 21 models are assumed equally-likely and were not selected from a set of more than 21 models. The different spatial patterns of $LMST$ underlying each of the 21 CMIP6 models reflect an important source of uncertainty within the model and, as such, we preserve and propagate this uncertainty throughout the integrated model. To evaluate the performance of the set of 21 models and our pattern scaling method we compare the resulting country-level temperatures against the historical time-series of global gridded data from Berkeley Earth (Berkeley Earth, 2024). Figure 5 overlays the historical $GMST$ observations from this dataset against the $GMST$ coming from FaIR. Figure 6 compares the result of using the country-level patterns discussed above to recover $CMST$ against the observed historical record from the gridded Berkeley Earth dataset. Both figures taken in tandem show that the $GMST$ and $CMST$ closely match the historical records, supporting the use of our pattern scaling approach.

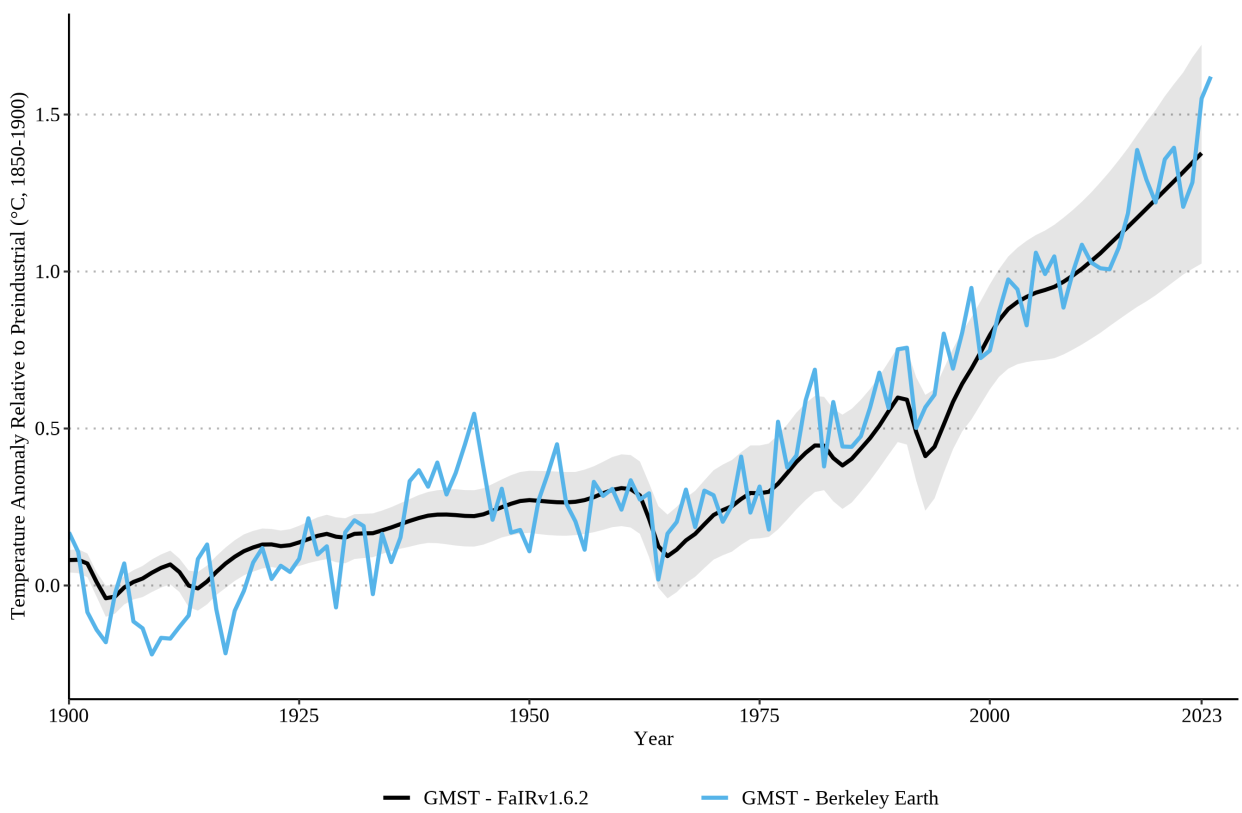


Fig. 5 | Historical Global Mean Surface Temperature ($GMST$) from FaIR and Berkeley Earth Black lines represent mean (solid) GMST from FaIR along with 1st to 99th percentile ranges (shaded ribbon) that are a result of the uncertainty underlying FaIR. Blue line represents the historical record from Berkeley Earth.

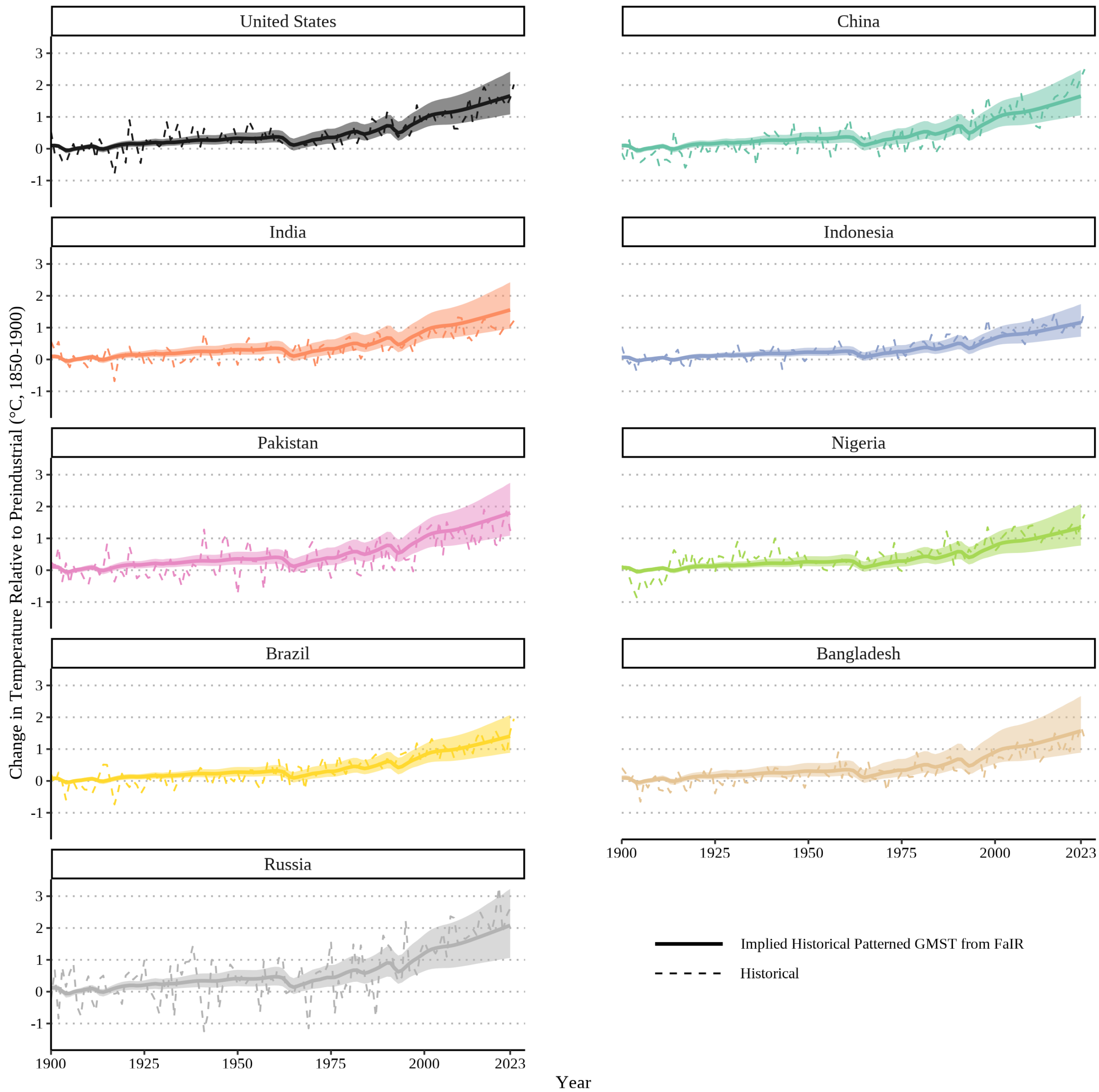


Fig. 6 | Historical Country-level Mean Surface Temperature ($CMST$) from Pattern Scaling Compared to Berkeley Earth Observations Solid lines represent mean surface temperature from combining FaIR GMST and population-weighted patterns ($\widetilde{\beta}_{sc} \times GMST_{st}$), along with 1st to 99th percentile ranges (shaded ribbons) that are a result of the joint uncertainty underlying FaIR and the CMIP6 patterns. Dashed lines represents the population-weighted historical record from Berkeley Earth.

2.4 SC-CO_2

As in [15] and [37], in each of the 10,000 Monte Carlo runs, we estimate climate damages for two cases: a 'baseline' case and a 'perturbed' case that adds an extra pulse of CO_2 emissions. We then calculate marginal climate damages in each time period t as the difference in modeled damages per ton between the pulse and baseline run:

$$\mathrm{MD}_t = \sum_{d=1}^{4} \sum_{r=1}^{R_d} \text{Damages with pulse}_{t,d,r} - \text{Baseline damages}_{t,d,r}, \tag{7}$$

where we aggregate over each of the four damage sectors d at their respective geographic resolutions r.

As [15] and [37], we estimate the 2020 SC-CO_2 by multiplying marginal damages by the stochastic discount factor SDF_t and aggregating across time:

$$\text{SC-CO}_2 = \sum_{t=2020}^{2300} \mathrm{SDF}_t \times \mathrm{MD}_t. \tag{8}$$

The stochastic Ramsey-like discount factor is defined in each Monte Carlo run r as

$$\mathrm{SDF}_t^r = \frac{1}{(1+\rho)^{t-2020}} \left(\frac{y_{t,avg}^r}{y_{2020,avg}} \right)^{-\eta} \tag{9}$$

where $y_{t,avg}^r$ is world average per capita income in year t in Monte Carlo run r and ρ is the pure rate of time preference. In the main body of the paper, we use $\rho = 0.2\%$, which is consistent with the main specification of [15] and [37]. In the extended data, we show a variety of alternative ρ values including Germany's preferred ρ value ($\rho = 1\%$) [31].

Equation 8 yields the SC-CO_2 in PPP dollars unadjusted for diminishing marginal utility across space as we show in figure 3, which values damages in low-income countries the same as a dollar of damages in high-income countries (what we refer to as "Equal Dollar Valuation"). This is the monetization approach used in EPA's 2023 SC-GHG report [15] and Rennert et al. 2022 [37]. We run this exercise in each of our 10,000 Monte Carlo runs that account for uncertainty in emissions, population, economic growth, and the response of the climate system. As in [15] and [37], we typically summarize the distribution of our 10,000 estimated SC-CO_2 by its mean, $\mathbb{E}[\text{SC-CO}_2]$, where the expectation operator is taken jointly over all the uncertain parameters determining $\mathrm{SDF}_{t,r}$ and MD_t.

2.4.1 The Income Weighted SC-CO_2

To estimate the income weighted SC-CO_2, we apply income weights to marginal damages following the U.S. Office of Management and Budget's (OMB's) update to Circular A-4 [32]. The updated document suggests constructing weights based on income (what it calls "income weights"). Following the guidance in [32], we multiply the marginal damages in each region in each time period, by the following weight:

$$w_{t,r} = \left(\frac{y_{t,r}}{y_{ref}}\right)^{-\eta} \tag{10}$$

where $y_{t,r}$ is the average income for region r in year t, y_{ref} is the reference point income level, and η is the elasticity of marginal utility. The income-weighted marginal damages are

$$\text{Income-Weighted MD}_t = \sum_{d=1}^{4}\sum_{r=1}^{R_d} w_{t,r}\text{MD}_{t,d,r} \tag{11}$$

As in [37], we run 10,000 Monte Carlo simulations to calculate 10,000 unique SC-CO_2 estimates. These sample the RFF-SP scenarios to account for uncertainties in emission trajectories, country-level population, and GDP growth levels; parametric uncertainties in the climate (FaIR) and sea level rise (BRICK) models; and damage function uncertainty in agriculture and temperature-related mortality.

OMB conducted a review of the literature on η as part of its update to Circular A-4 (see section 9.d of [95] for details). This review extended Acland and Greenberg's 2023 review [96]. Acland and Greenberg tentatively concluded using $\eta = 1.6$ with lower and upper bound sensitivity testing at 1.2 and 1.6. The 2023 Circular A-4 determined that $\eta = 1.4$ is a reasonable estimate for income weighting in regulatory analysis. We thus use income weighting with $\eta = 1.4$ in our main specifications in the main text. Likewise, the U.K. Green Book also suggests constructing weights based on income (what it calls "distributional weights") using an income elasticity of marginal utility of $\eta = 1.3$ [30]. The German government uses $\eta = 1$ to calculate the SC-CO_2. We show our results with the U.K.'s and Germany's preferred η in the supplementary data.

We can derive the weight in equation 10 by considering a standard isoelastic utility function that uses the income elasticity of marginal utility parameter η (we also refer to this as the utility curvature parameter):

$$u(y) = \frac{y^{1-\eta}}{1-\eta} \tag{12}$$

We use the standard social welfare function (SWF) used in the income weighting literature [20, 18] that aggregates utilities across a population of N individuals across time and discounted to present value using the pure rate of time preference parameter ρ discussed above:

$$W = \sum_{t=2020}^{2300} \sum_{i=1}^{N_t} u(y_{t,i}) \frac{1}{(1+\rho)^{t-2020}} \tag{13}$$

The utility-denominated SC-CO_2, which we'll label as SC-CO_2-u, is equivalent to the marginal damage caused by a marginal pulse of SC-CO_2 emissions:

$$\text{SC-CO}_2\text{-u} = \frac{\partial W}{\partial E} = \frac{\partial \sum_{t=2020}^{2300} \sum_{i=1}^{N_t} u(y_{t,i}) \frac{1}{(1+\rho)^{t-2020}}}{\partial E} \tag{14}$$

Which is equivalent to:

$$\text{SC-CO}_2\text{-u} = \sum_{t=2020}^{2300} \sum_{i=1}^{N_t} \frac{\partial u(y_{t,i})}{\partial E} \frac{1}{(1+\rho)^{t-2020}} \tag{15}$$

Applying the chain rule:

$$\text{SC-CO}_2\text{-u} = \sum_{t=2020}^{2300} \sum_{i=1}^{N_t} \frac{\partial u(y_{t,i})}{\partial y_{t,i}} \frac{\partial y_{t,i}}{\partial E} \frac{1}{(1+\rho)^{t-2020}} \tag{16}$$

The term $\frac{\partial y_{t,i}}{\partial E}$ represents the marginal damage to person i in time t from the emission E, so we can replace that term with $MD_{t,i}$:

$$\text{SC-CO}_2\text{-u} = \sum_{t=2020}^{2300} \sum_{i=1}^{N_t} \frac{\partial u(y_{t,i})}{\partial y_{t,i}} \frac{1}{(1+\rho)^{t-2020}} MD_{t,i} \tag{17}$$

No integrated assessment models used to calculate the SC-CO_2 are able to project damages at the level of individuals, so we instead represent damages at the aggregated region level r:

$$\text{SC-CO}_2\text{-u} = \sum_{t=2020}^{2300} \sum_{r=1}^{R} \frac{\partial u(y_{t,r})}{\partial y_{t,r}} \frac{1}{(1+\rho)^{t-2020}} MD_{t,r} \tag{18}$$

where $y_{t,r}$ represents the average income of region r at time t. This SC-CO_2 in equation 18 is in units of utility. Indeed, this can be used directly in BCA, but this tends to be inconvenient because other benefits and costs are usually measured in dollars [18]. For practical application in BCA, this utility-denominated SC-CO_2 must be converted into dollars. When income weighting, this is done with the following inverse marginal utility weight:

$$\mu_{ref} = \frac{1}{\frac{\partial u(y_{ref})}{\partial y_{ref}}}. \tag{19}$$

μ_{ref} represents the inverse marginal utility of some person with the reference point income level y_{ref}. Thus, this factor converts the SC-CO_2 from units of utility into units of money as it is valued by someone with the reference point level of income:

$$\text{SC-CO}_2 = \sum_{t=2020}^{2300} \sum_{r=1}^{R} \mu_{ref} \frac{\partial u(y_{t,r})}{\partial y_{t,r}} \frac{1}{(1+\rho)^{t-2020}} MD_{t,r} = \sum_{t=2020}^{2300} \sum_{r=1}^{R} \frac{\frac{\partial u(y_{t,r})}{\partial y_{t,r}}}{\frac{\partial u(y_{ref})}{\partial y_{ref}}} \frac{1}{(1+\rho)^{t-2020}} MD_{t,r} \tag{20}$$

Now, the marginal damage in time t in region r ($MD_{t,r}$) is multiplied by the ratio of the marginal utility for an average person in that region at that time ($\frac{\partial u(y_{t,r})}{\partial y_{t,r}}$) and the marginal utility at the reference point income level ($\frac{\partial u(y_{ref})}{\partial y_{ref}}$). Because of diminishing marginal utility, marginal utility for people at lower income levels is higher (i.e., a dollar is more valuable to someone with lower income), whereas the marginal utility for those at higher income levels is lower (i.e., a dollar is less valuable to someone with higher income). Thus, equation 20 shows mathematically how income weighting upweights marginal damages in lower-income regions (because $\frac{\partial u(y_{t,r})}{\partial y_{t,r}}$ in the numerator is larger) and downweights marginal damages in higher-income regions (because $\frac{\partial u(y_{t,r})}{\partial y_{t,r}}$ is smaller).

Furthermore, equation 20 illustrates the importance of the reference point income level. When one chooses a higher reference point income level (e.g., average income in the U.S.), this increases the SC-CO_2 because $\frac{\partial u(y_{ref})}{\partial y_{ref}}$ (in the denominator) is lower. Whereas if one chooses a lower reference point income level (e.g., average income in India), this decreases the SC-CO_2 because $\frac{\partial u(y_{ref})}{\partial y_{ref}}$ is higher. This mathematically illustrates the intuition that we mentioned in the main text: using a higher-income reference region increases the SC-CO_2 just by virtue of people in that region valuing money less. Whereas using a lower-income reference region decreases the SC-CO_2 just by virtue of people in that region valuing money more.

We can simplify equation 20 further by assuming isoelastic utility (plugging in equation 12):

$$\text{SC-CO}_2 = \sum_{t=2020}^{2300} \sum_{r=1}^{R} \frac{y_{t,r}^{-\eta}}{y_{ref}^{-\eta}} \frac{1}{(1+\rho)^{t-2020}} MD_{t,r} = \sum_{t=2020}^{2300} \sum_{r=1}^{R} \left(\frac{y_{t,r}}{y_{ref}}\right)^{-\eta} \frac{1}{(1+\rho)^{t-2020}} MD_{t,r} \quad (21)$$

Now, we can see that the the term is $\left(\frac{y_{t,r}}{y_{ref}}\right)^{-\eta}$ is exactly equal to the weight given in equation 10.

2.4.2 Comparing the SC-CO_2 Before and After Income Weighting

We can compare the expression for the income-weighted SC-CO_2 in equation 21 with the SC-CO_2 before income weighting in equation 8. Writing out the full expression for the SC-CO_2 before income weighting from equation 8, including the SDF:

$$\text{SC-CO}_2 = \sum_{t=2020}^{2300} \frac{1}{(1+\rho)^{t-2020}} \left(\frac{y_{t,avg}}{y_{2020,avg}}\right)^{-\eta} \text{MD}_t. \quad (22)$$

Since the term $\left(\frac{1}{y_{2020,avg}}\right)^{-\eta}$ is a constant—recall that $y_{2020,avg}$ is simply world average per capita income in year 2020—we can move it to the front of the expression. Since MD_t is just the sum of marginal damage across all regions in time t, we can rewrite this expression as:

$$\text{SC-CO}_2 = (y_{2020,avg})^{\eta} \sum_{t=2020}^{2300} \frac{1}{(1+\rho)^{t-2020}} (y_{t,avg})^{-\eta} \sum_{r=1}^{R} MD_{t,r} \tag{23}$$

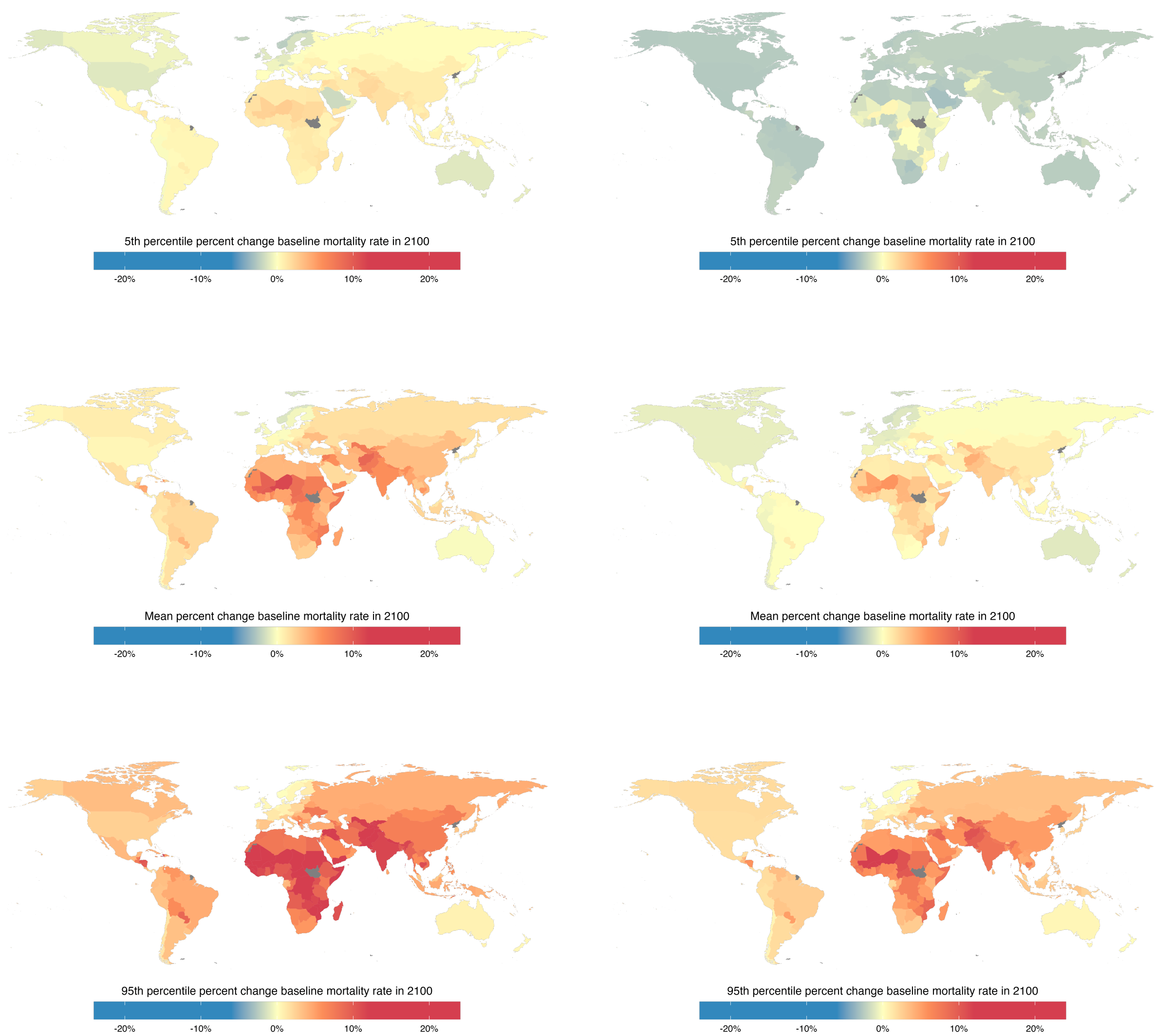
Now consider a special case of the income-weighted SC-CO_2 (equation 21) where global average income in 2020 is the reference point income level (the reference point income level that we used in multiple results in the main text):

$$\text{SC-CO}_2 = (y_{2020,avg})^{\eta} \sum_{t=2020}^{2300} \frac{1}{(1+\rho)^{t-2020}} \sum_{r=1}^{R} (y_{t,r})^{-\eta} MD_{t,r} \tag{24}$$

To summarize, equation 23 is the SC-CO_2 before income weighting, and equation 24 is the SC-CO_2 after income weighting (using global average income as the reference point, which is the most directly comparable to the SC-CO_2 before income weighting). These results show the similarities between the SC-CO_2 before and after income weighting. The only difference between these two equations is that after income weighting (equation 24), the marginal damage in each region in each time period is multiplied by the marginal utility in that region at that time, $(y_{t,r})^{-\eta}$, while before income weighting (equation 23), marginal damages are first aggregated across all regions in each time period and then multiplied by the global average marginal utility $(y_{t,avg})^{-\eta}$. Thus, the income-weighted SC-CO_2 (equation 24) accounts for differences in marginal utility across time and across regions, whereas the SC-CO_2 before income weighting (equation 23) does not account for differences in marginal utility across regions (but does account for global average differences in marginal utility across time.)

A Supplementary Data

(a) Without income-based adaptation **(b) With income-based adaptation**



Extended Data Fig. 1 | The spatial distribution of projected mortality impacts in 2100. The first row shows the 5th percentile of damages across the 10,000 Monte Carlo simulations; the middle rows the mean damages across the 10,000 Monte Carlo simulations, and the bottom row shows the 95th percentile across the Monte Carlo simulations. We take averages across the 10,000 draws in a Monte Carlo simulation, which captures uncertainty in socioeconomic and emissions scenarios (RFF-SPs), uncertainty in climate (FaIR v1.6.2), and damage function parameters (see methods section for details). These results only represent temperature-related mortality and do not account for other climate-mortality pathways. **(a)** Shows results without accounting for income-based adaptation, i.e., it assumes that historical levels of adaptation are simply preserved in the future, and more affluent populations will be just as vulnerable to heat as historically observed populations. **(b)** Shows results accounting for income-based adaptation.

Table A.1: The 2020 $SC\text{-}CO_2$ under different parameter values

Specification	Equal Dollar Value $SC\text{-}CO_2$	Income-Weighted $SC\text{-}CO_2$
$\eta = 1.4, \rho = 0.2\%$ (main text)	218	527
	[-50 - 690]	[61 - 1,354]
$\eta = 1.4, \rho = 0.1\%$	244	588
	[-58 - 786]	[60 - 1,531]
$\eta = 1.4, \rho = 0.0\%$	275	661
	[-70 - 892]	[60 - 1,746]
$\eta = 1.24, \rho = 0.2\%$ (EPA 2023 SC-GHG report)	233	496
	[-80 - 728]	[36 - 1,238]
$\eta = 1, \rho = 1\%$ (Germany preferred)	124	224
	[-49 - 349]	[21 - 500]

Table shows mean and 5th–95th quantile 2020 $SC\text{-}CO_2$ values across 10,000 Monte Carlo runs using various utility curvature parameters and pure rates of time preference. Income-weighted values use global average income as the reference point.

Table A.2: The 2025 $SC\text{-}CO_2$ Using Each Country's Average Income as the Reference Point

Country	$SC\text{-}CO_2$	Country	$SC\text{-}CO_2$	Country	$SC\text{-}CO_2$	Country	$SC\text{-}CO_2$
Afghanistan	28	Djibouti	410	Lesotho	42	Samoa	414
Albania	382	Dominican Republic	575	Liberia	24	Saudi Arabia	3,272
Algeria	555	Ecuador	399	Libya	428	Senegal	69
Angola	73	Egypt	404	Lithuania	1,017	Serbia	570
Argentina	698	El Salvador	212	Luxembourg	6,474	Sierra Leone	32
Armenia	347	Equatorial Guinea	1,076	Macao SAR China	7,922	Singapore	4,754
Aruba	1,262	Eritrea	19	Madagascar	28	Slovakia	1,141
Australia	3,145	Estonia	1,261	Malawi	18	Slovenia	1,543
Austria	2,552	Eswatini	267	Malaysia	1,130	Solomon Islands	171
Azerbaijan	820	Ethiopia	34	Maldives	639	Somalia	14
Bahamas	1,167	Fiji	534	Mali	54	South Africa	467
Bahrain	4,055	Finland	2,194	Malta	1,780	South Korea	1,984
Bangladesh	88	France	2,132	Mauritania	84	Spain	1,706
Barbados	1,385	French Polynesia	1,769	Mauritius	978	Sri Lanka	474
Belarus	511	Gabon	731	Mexico	678	St. Lucia	786
Belgium	2,211	Gambia	37	Moldova	316	St. Vincent & Grenadines	578
Belize	219	Georgia	348	Mongolia	513	Sudan	108
Benin	46	Germany	2,848	Montenegro	573	Suriname	551
Bhutan	138	Ghana	146	Morocco	268	Sweden	2,765
Bolivia	192	Greece	1,020	Mozambique	19	Switzerland	4,306
Bosnia & Herzegovina	469	Guatemala	231	Myanmar (Burma)	161	Syria	68
Botswana	688	Guinea	43	Namibia	358	São Tomé & Príncipe	149
Brazil	588	Guinea-Bissau	33	Nepal	54	Taiwan	2,482
Brunei	5,062	Guyana	607	Netherlands	2,807	Tajikistan	59
Bulgaria	715	Haiti	35	New Caledonia	1,475	Tanzania	56
Burkina Faso	28	Honduras	114	New Zealand	1,948	Thailand	607
Burundi	12	Hong Kong SAR China	2,937	Nicaragua	150	Timor-Leste	74
Cambodia	84	Hungary	1,152	Niger	14	Togo	28
Cameroon	80	Iceland	2,713	Nigeria	143	Tonga	537
Canada	2,754	India	185	North Macedonia	453	Trinidad & Tobago	1,409
Cape Verde	275	Indonesia	386	Norway	6,298	Tunisia	393
Central African Republic	9	Iran	777	Oman	2,497	Turkey	1,181
Chad	36	Iraq	805	Pakistan	133	Turkmenistan	658
Chile	1,043	Ireland	4,540	Palestinian Territories	145	Uganda	39
China	544	Israel	1,706	Panama	1,027	Ukraine	239
Colombia	533	Italy	1,883	Papua New Guinea	212	United Arab Emirates	1,678
Comoros	64	Jamaica	235	Paraguay	293	United Kingdom	2,201
Congo - Brazzaville	168	Japan	2,315	Peru	450	United States	3,567
Congo - Kinshasa	14	Jordan	349	Philippines	233	Uruguay	923
Costa Rica	563	Kazakhstan	1,283	Poland	1,284	Uzbekistan	98
Croatia	951	Kenya	70	Portugal	1,195	Vanuatu	177
Cuba	765	Kuwait	2,062	Puerto Rico	1,383	Venezuela	614
Cyprus	1,668	Kyrgyzstan	55	Qatar	3,342	Vietnam	177
Czechia	1,219	Laos	212	Romania	977	Yemen	56
Côte d'Ivoire	79	Latvia	1,015	Russia	1,308	Zambia	87
Denmark	2,819	Lebanon	650	Rwanda	38	Zimbabwe	50

Table shows mean 2025 $SC\text{-}CO_2$ values in dollars across 10,000 Monte Carlo simulations for each country's income as the reference point. The choice of reference point does not affect projected damages; rather, it changes the units in which these damages are represented. The table uses a pure rate of time preference $\rho = 0.2\%$ (consistent with the main specification of [15] and [37]) and a utility curvature parameter of $\eta = 1.4$ (which reflects the OMB BCA guidelines from 2023 [32]).

Software

All our results are computed using open-source software tools. We use the Julia programming language for the modeling [97]. All models used in this study are implemented on the Mimi.jl computational platform for integrated assessment models [98]. Figures were produced using Julia and R.

Data Availability

Complete replication code and data for this study will be made public in a public GitHub repository and registered with Zenodo alongside publication.

Code Availability

Complete replication code and data for this study will be made public in a public GitHub repository and registered with Zenodo alongside publication.

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Competing Interests

The authors have no competing interests to report.